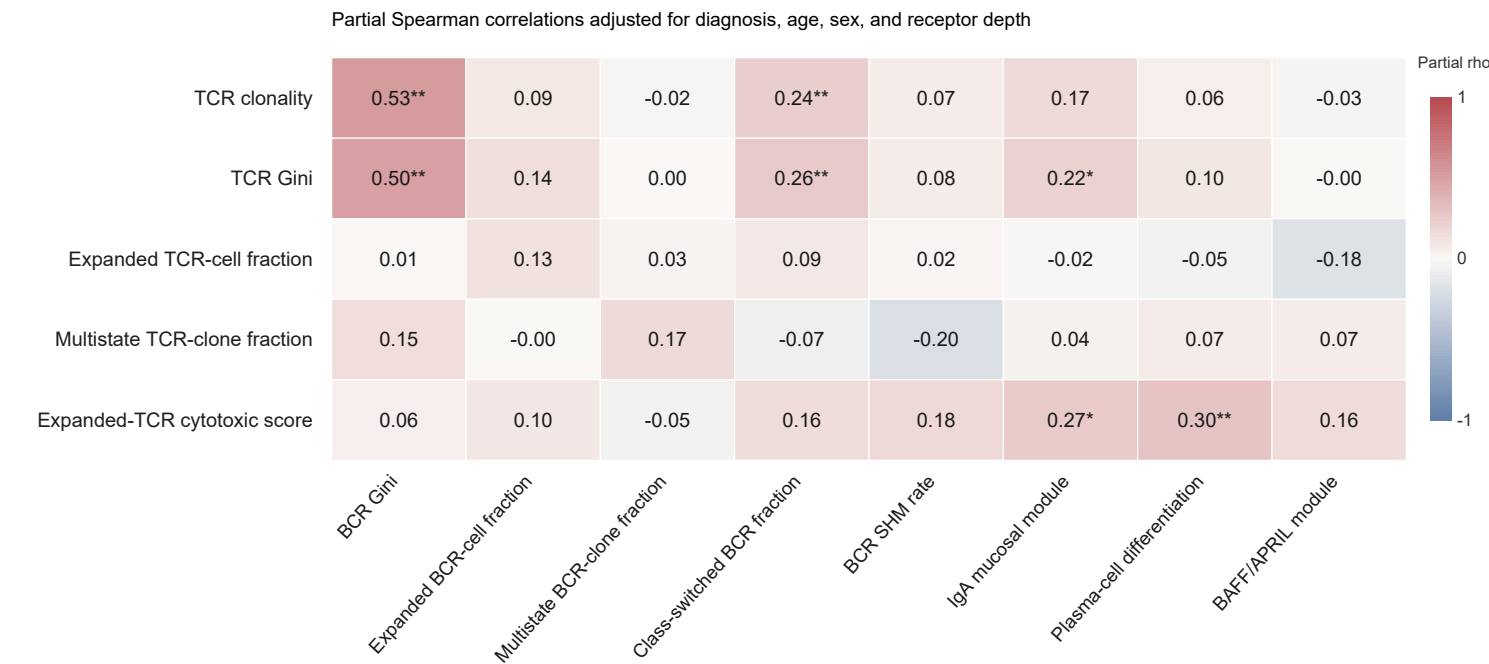
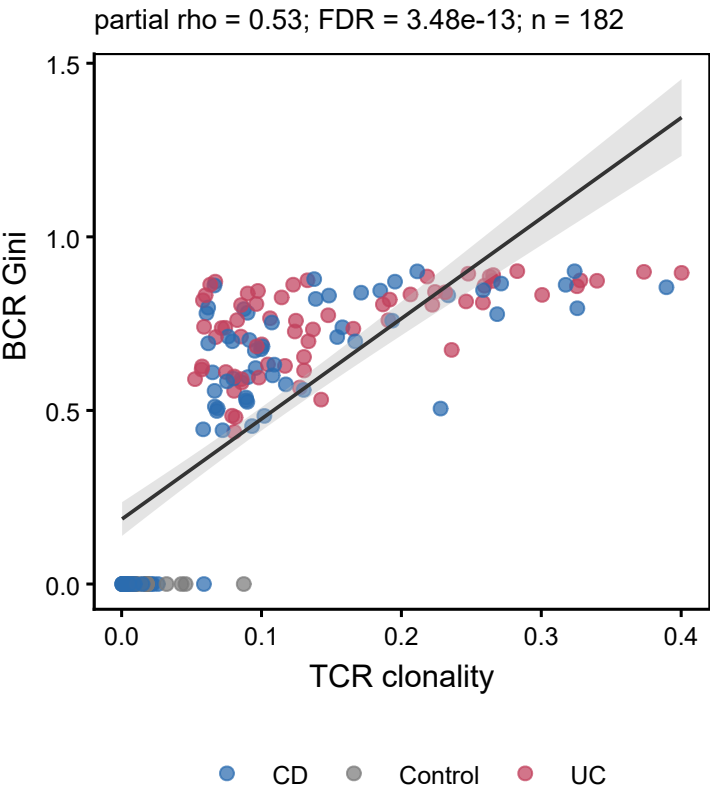


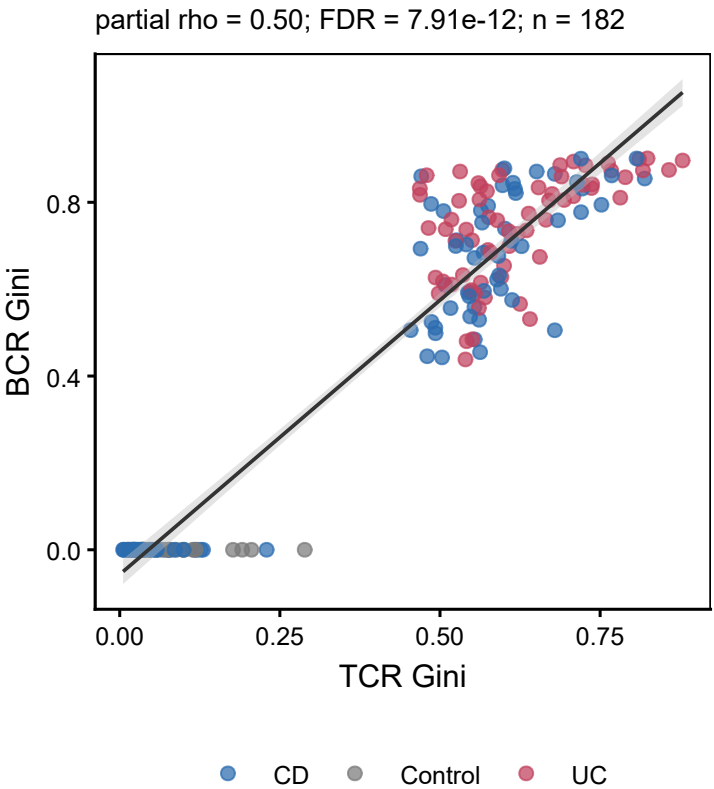
A Participant-level TCR-BCR covariation



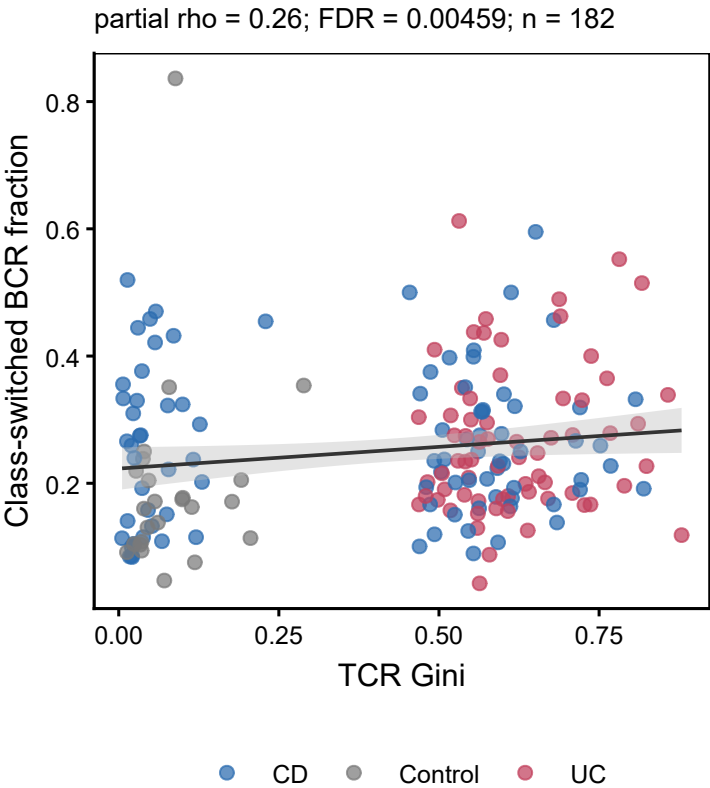
B TCR clonality versus BCR Gini



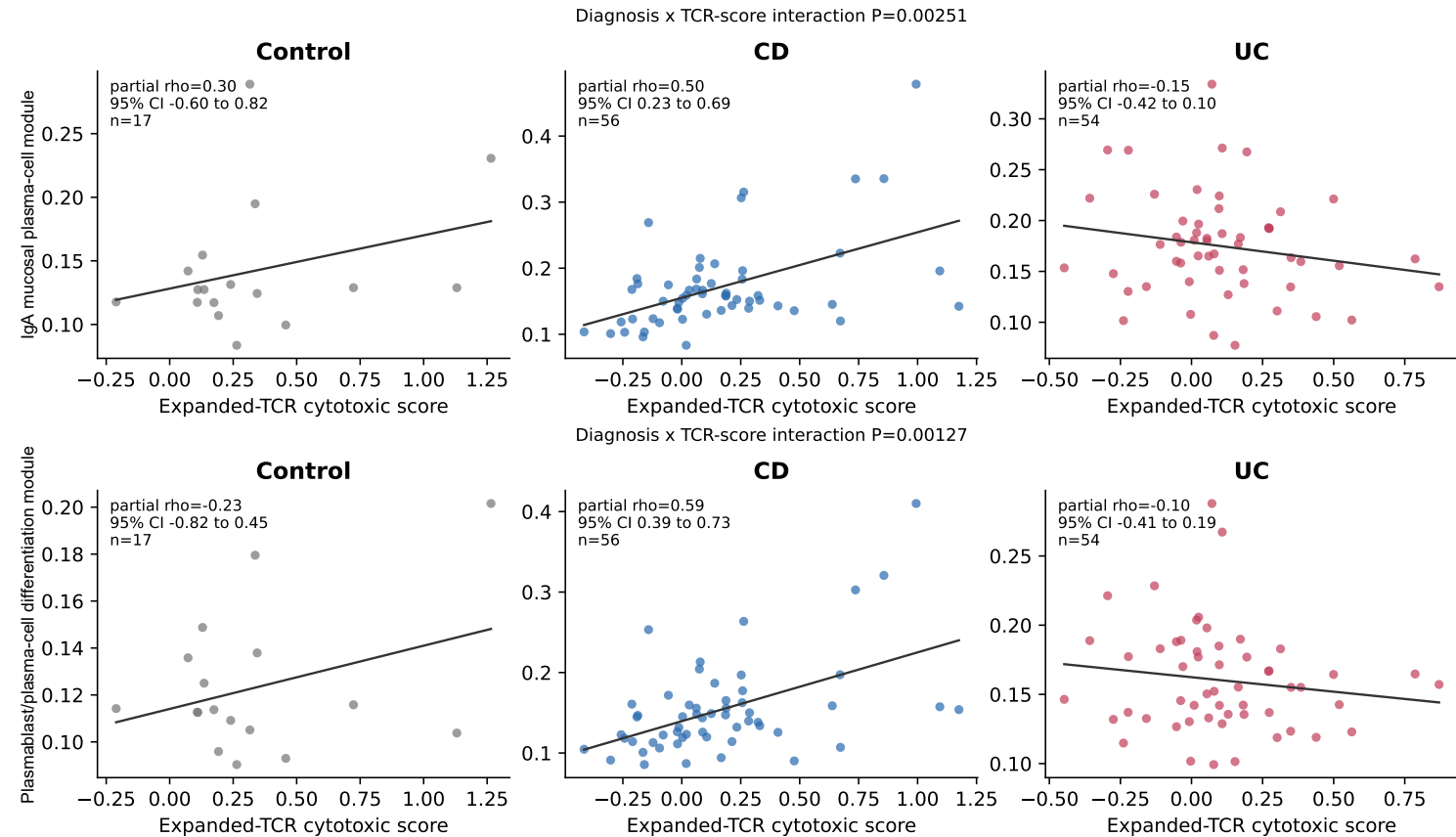
C TCR Gini versus BCR Gini



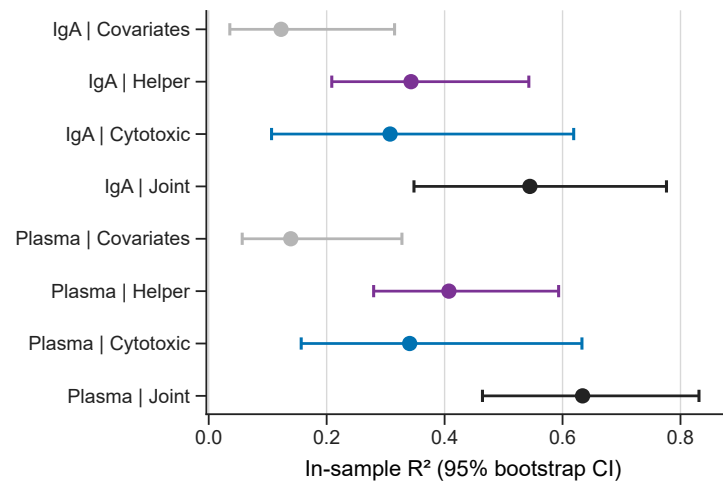
D TCR Gini versus class-switched BCR fraction



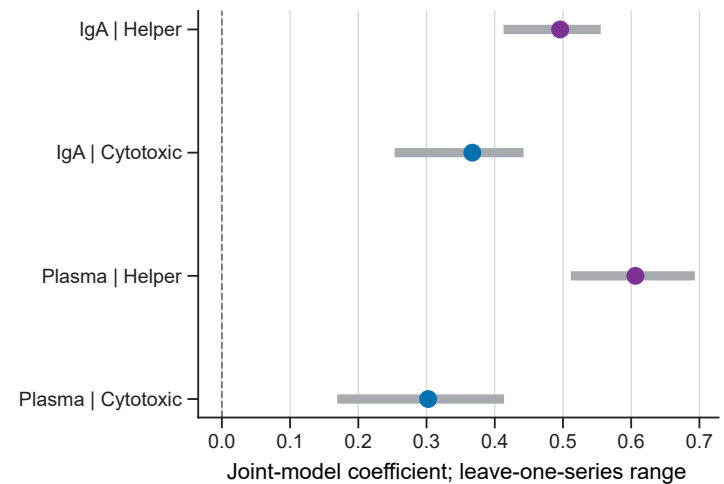
E Cytotoxic T-cell and plasma-cell programs covary selectively in Crohn's disease



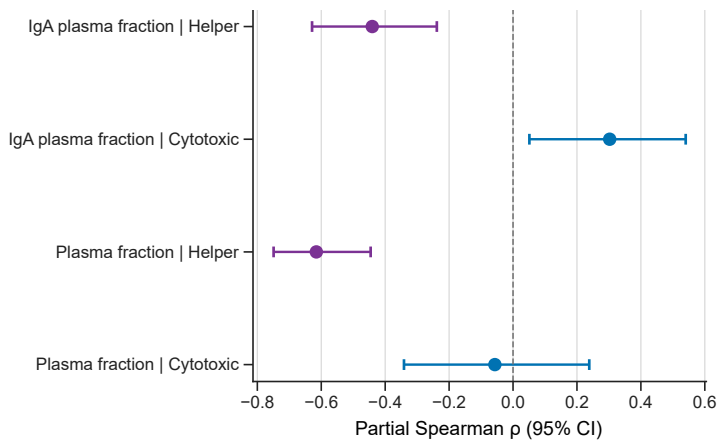
F Model variance and incremental information



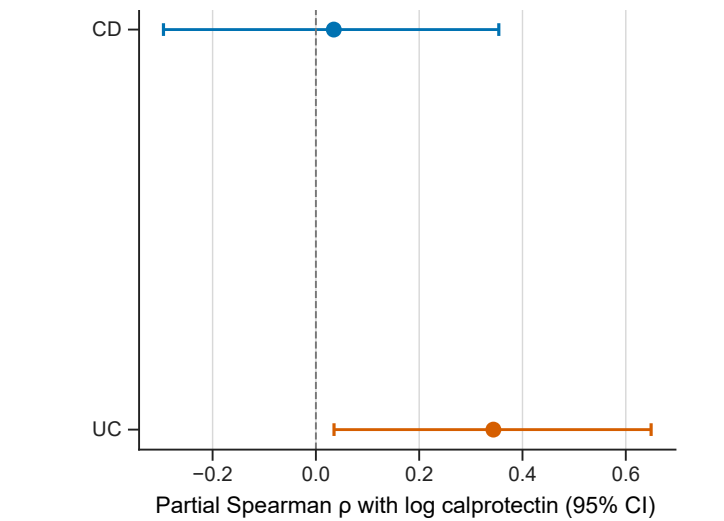
G Acquisition-series influence



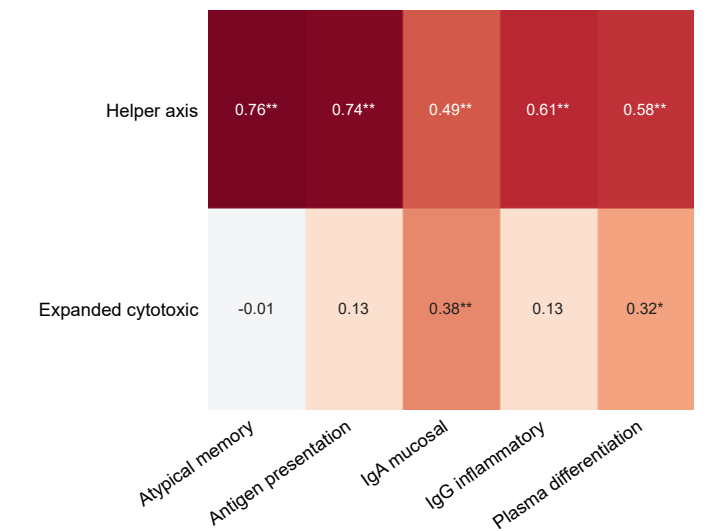
H Orthogonal cell-composition validation



I Contemporaneous clinical association



J Cross-program specificity in Crohn's disease



K Formal between-program contrasts

